

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction,for effective data preparation (BL3,BL4) Assessment Tool Weightage: 20%

QUESTIONS: 10

Total Marks:20

Annexure A

CLASS TEST

Criteria

Understanding Procedure of Task (2)

Observation | Viva Voce / & Execution & Result (2) (2.5)

Conceptual
Response (2)

Time Management

Total(10)

& Presentation

(1.5)

25%

20%

20%

15%

100%

Weightage 20%

02
93
04
95
96
07
Q8
09
Q10

Code of Conduct:

1

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*(Name / Reg No) certify that this submission is
my original work*

and that I have adhered to the guidelines specified for this assessment.

*I understand that any violation of **academic** integrity rules will result in disciplinary
action.*

Signature of the student:

CLASS TEST

QUESTIONS:

Total Marks: 10x2=20

Q1. Define Data Mining and list any two basic data mining tasks.

Explain the steps involved in KDD process and justify the role of data preprocessing in it. [L2, L4]

Q2. **What** is data cleaning? Mention two common data quality **issues**. **Explain methods to** handle missing values and noisy data with **suitable examples**. [L1, L2, L3]

Q3. Define **data** integration and list two challenges **associated with it**. **Explain how** redundancy **and** inconsistency are handled during data **integration**. [L2, L4]

Q4. **What is covariance** analysis? List the two covariance formalms [L2, L3]

Q5. Suppose two stocks A and B have **the** following values in **one week**: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). **Question**: If **the** stocks are affected by **the same industry trends**, will their prices rise or **fall together and independent** ?

Q6. **Explain the** architecture **of** a typical data mining system **with** a neat **diagram**. Discuss **the role of each component** and analyze how **they** interact to support **knowledge discovery**. [L2, LA]

Q7. Find **covariance** for following data set $x = \{4, 5, 6, 7, 9\}$, $y = \{8, 3, 7, 5, 6\}$

Q8. **Describe the major** steps in data pre-processing. Given a **dataset** with **missing values, outliers, and inconsistent** formats, propose a **systematic pre-processing pipeline** and justify the **sequence of steps**. [L3, L4, L5]

Q9. A **hospital is analyzing** patient age data to identify broad **health trends**. **Because** the **raw data contains minor variations and noise**, the analyst needs to **perform** local smoothing using equal-frequency (**equi-depth**) binning. **Sample Data Set (Ages)**: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4]

Q10. Explain **data discretization** and its importance in data mining. Compare equal-width **and** equal-frequency binning with **suitable examples**, and evaluate **their** impact on data analysis. [L3, L4, 15]

Define Data

Mining and list

any

two

basic

data mining tasks. Explain the steps involved

in it.

Data mining is the process of discovering
interesting previously unknown, and
potentially

useful patterns from large volumes of data

Statistical mathematical and machine

Q..

mining

learning.

Turo basic tasks:-

1

пови́тся

Alvindication - Arigning doser to
Paredefineet Chestering Grouping
Similar data objects togethon
without predefined labels.

Koo Procues Steps:-

1.) **Data** Meaning 2.) Dala
Integration 3.) Daha Selection. 4.) Data
transformation 5.) Data Mining 6.)
Patterns Exaluation 7.) knowledge
Presentation.
Daha Cleaning and
Issues:-

incomplete

09

Definition: Fining
diarty data

Chemovin
g

inaccurate,

Quality Issues:- Miciary valued, noisy
data Handling Missing:- Ignore Record,
fill with mean

medion on

Constant.

thandling Noise- Bining, Pugnacion
fitting

lonorciame Analysis and Formula's

Definition: Measure

direction of linear

relationship b/co

2

2

Variables

Sample: Bay = { (x; - x)
(y; - y)

Population: Jay

бжу

m-1

3-1

y

$\Sigma (m; - Me) (Y) -$
My)
Cai

Q.5.) Shock lom-relation

Problem:

47 2014

Result:- Cove (A,B)

2

= +5.0

A

2+3+5+4+6

5

= 4

3

$$= 5 + 8 + 10 + 11 + 14$$

Condurion : Positive
Comariame

$$\begin{aligned} \cdot \text{Cov} (A, B) &= (-2) (-4 \cdot 6) + (-1) (-1) + \\ & (1) (0.4). \\ & ((0) (1.4) + (2) (4.4). \end{aligned}$$

=2014=

(26) Data

Data Mining Bystem
Architekures

Flow- Data Source DBY Warehous
→

Завис

Doctor Mining Engine - Patton
Exclusion

User Interface

I

love Role : – Mimary Enging
extrace

Finaliation engine drops comenterating

Datsune:

ones

Crimern

$$x = \{4, 5, 6, 7\}$$

$$4, 5, 6, 7, 93$$

$$9 =$$

$$24.3$$

7

Means →

$$x = 4 + 5 + 6 + 7 + 9$$

$$\tilde{y} =$$

1

3

$$8 + 3 + 7 + 5 + 6$$

Sum of
Products:-

-

5

$$6.2$$

=

$$-4.84 + 3.36 - 0.24 - 0.64 + 0.56$$

$$5.8$$

-0.45

Reduce

Bin 3. (9 items):- 35, 35, 35, 36, 40, 45, 46,
Mean: 43.78

52,70 →

Discretization And Bining Comparison

Data

Discretization:- Converting
Continuous

data into discrete / Categorical

number into

Equal Width :-

intervals.

Divide Range into equal

intervals to Cutliers,

Distance :-

exactly empty bins.

o

Divides data

Equal

Frequency

his frome

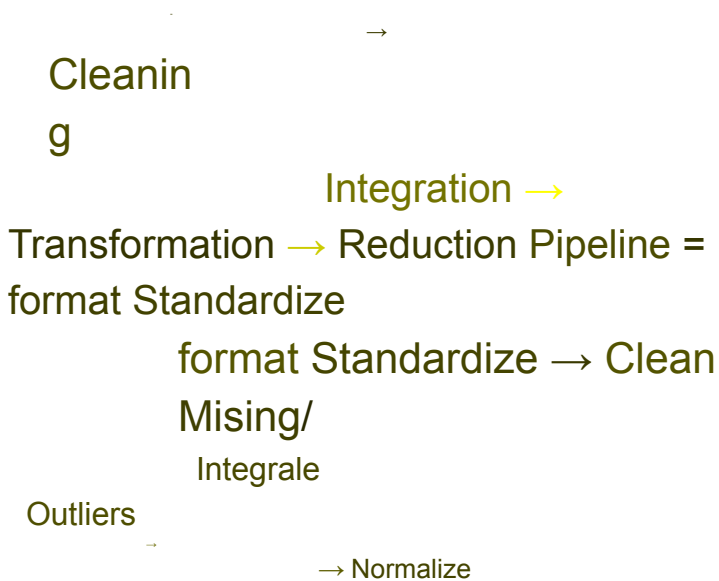
Count, bin
Admontag
e
well.
80 lach

handles skemed
distributions

Sample Covariance
(Say):.

Say
-1.80
5-1

Qs) Pane proceering steps
and Pipeline



Justification:- Fin Outliers
before Saling Scale before
reducing nolume.

89.) Equal Frequency Binning

(24 Arges, 3 Bine)

Bin 1 (a itcoms) : 13, 15, 16, 16, 19, 20, 20, 21, 22 → Mean = 18.21122

Bin 2 (9 items) : 22, 25, 25, 25, 25, 30, 33, 33, 35
Mean = 28.11